
MB 550 : Artificial Intelligence for Marketing

Course Code : MB 550

Course Name : Artificial Intelligence for Marketing

L-T-P-C : 2-0-0-2

Intended for : MBA

Prerequisite :

Mutual Exclusion : Organizational Management HS304 offered by SHSS

Approval: 54th BoA

Course Contents

- Foundations of AI: Refreshing Concepts of Machine Learning (ML) Methods- Supervised, Unsupervised, and Reinforcement Learning; Maximum Likelihood Estimation, Principal Component Analysis and Clustering, Logistic Regression and Multinomial Classification, and Naive Bayes Classifiers; AI Types, AI and Algorithmic Marketing, Defining AI and Algorithmic Marketing, Marketing functions Automation and Augmentation, Why AI for Marketing? (4 Hours)
- AI for Predictive Modelling: Business Objectives, Consumer Choice Theory- Multinomial Logit Model, Survival Analysis. (4 Hours)
- AI for Finding Optimal Match Between Customers and Offerings: Fundamentals of Product Discovery Problem and identifying right customers for a given offering, Promotions and Advertisements- Business Objectives, Targeting Pipeline, Response Modelling and Measurement, Targeting and LTV Models- Propensity Modelling, Segmentation and Persona-based Modelling, Targeting by using Survival Analysis, Lifetime Value Modelling, Markov Chain Models, Designing and Running Campaigns, Online Advertisements, Measuring the Effectiveness. (5 Hours)
- AI Helps Finding Products for the Customers- Search: Business Objectives, Matching and Ranking- Token Matching, Normalization and Stemming, Ranking and the Vector Space Model; Semantic Analysis, Latent Semantic Analysis, Word2Vector Model, Search Methods for Merchandising. (3 Hours)
- Recommending Products for the Customers: Business Objectives, Quality Evaluation, Recommendation Methods- Content-based, Collaborative Filtering, and Model-based Collaborative Filtering, Contextual Recommendations, Non-Personalized Recommendations. (4 Hours)
- Pricing and Assortment: Business Objectives, The Impact of Pricing, Price and Value, Price and Demand, Basic Price Structures, Demand Prediction, Price Optimization, Dynamic Pricing, Store-Layout Optimization and Category Management. (4 Hours)
- ChatBots and Large Language Models (LLMs) for Marketing: Business Objectives, Bots as a New Customer Interface and Operating System, Harnessing the Power of LLMs (like, ChatGPT) for Your Business, LLMs for Lead Generation, social media marketing, Optimizing Conversion Rates, Market Research and Analysis. (4 Hours)

Textbooks:

1. NA

References:

1. Jim Sterne, **Artificial Intelligence for Marketing: Practical Applications**, John Wiley & Sons, 2017.
2. Mike Kaput, and Paul Roetzer, **Marketing Artificial Intelligence: AI, Marketing, and the Future of Business**, BenBella Books, 2022.
3. Ilya Katsov, **Introduction to Algorithmic Marketing: Artificial Intelligence for Marketing Operations**, Ilia Katcov, 2017.
4. Raj Venkatesan, and Jim Lecinski, **The AI Marketing Canvas: A Five-Stage Road Map to Implementing Artificial Intelligence in Marketing**, Stanford Business Books, 2021.
5. Peter Gentsch, **AI in Marketing, Sales and Service: How Marketers without a Data Science Degree can use AI, Big Data and Bots**, Springer Nature, 2019.
6. Mike Kapu, Paul Roetzer, **Marketing Artificial Intelligence: AI, Marketing, and the Future of Business**, BenBella Books, 2022.
7. Katie King, **Using Artificial Intelligence in Marketing: How to Harness AI and Maintain the Competitive Edge**, Kogan Page, 2019.